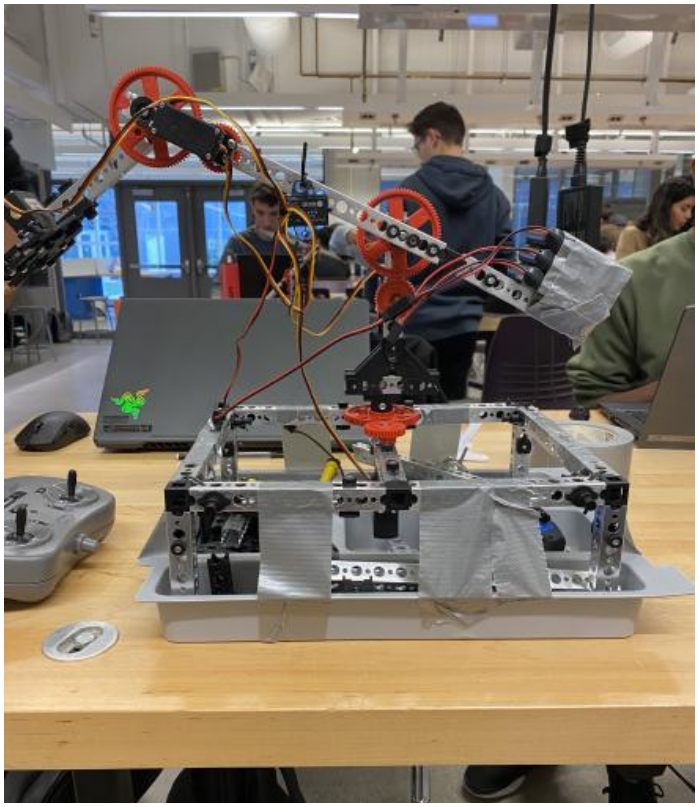
Mechanical Projects - Karthigan U



Highly Simplified Surgery Robot

Objective: Pick up 4 different bones and place them in the the model humans body, without dropping them or being too aggressive - made in a team of 4

Competition Restrictions:

- Only given 2 90 degree servos, and 1 continuous
- No human interference (results in point deduction
- Only allowed to use the materials given in the box

Our Approach:

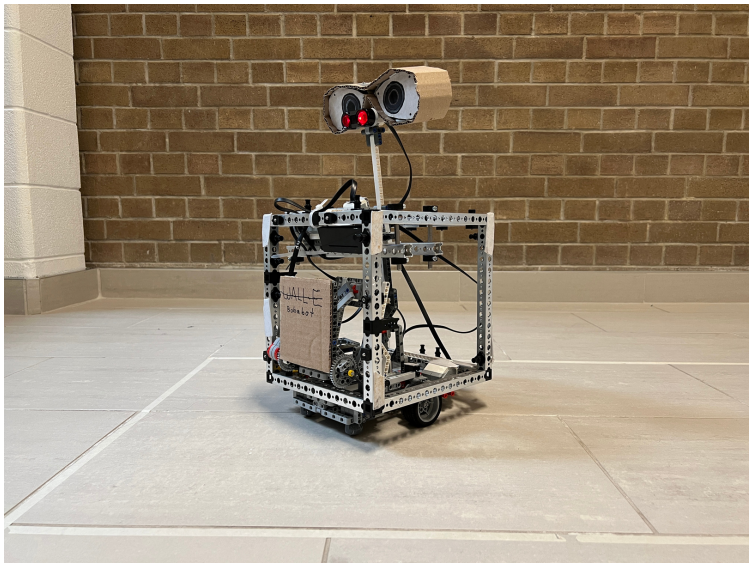
- Used a claw mechanism,
  - 90 degree servo to open and close the claw
  - 90 degree servo to rotate arms bas e
  - continuous servo use as a pivot to gain maximum height
- Used tray given in the box to hold parts (added stability to base)
- Taped dead batteries to the opposite end of arm from the claw (acted as counterbalance

Results: Received 124/125 possible points in the demonstration

BobaBot - A Drink Delivery Robot

Robot Tasks

- Prompt user for desired table number and drink size
- Open the drink holder and receive drink from user
- Navigate to desired table while avoiding obstacles
- Process customer payment
- Deliver drink (based on if customer has sufficient funds in their card to pay the balance)
- Return to starting position
- Update card database and display updated amount of money in accounts

Results

- Robot successfully completed all its tasks with minimal drift from it's desired pathway (finished within 3cm of starting location)